
Patient-Readable Clinical Trial Descriptions: A Framework for Summarization and Text Simplification

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Abstract

Understanding clinical trial descriptions is challenging for patients and the general public due to technical medical language and lengthy, complex narratives. Although public registries such as ClinicalTrials.gov (4) aim to promote transparency, many trial descriptions exceed recommended readability levels for patient-facing materials, limiting informed decision-making and participation. We propose an end-to-end, patient-centered framework that uses large language models to transform clinical trial records into clear, accessible summaries. Our pipeline leverages both structured metadata and unstructured text, integrating controlled summarization, linguistic simplification, medical jargon identification, and glossary-based explanations. We explore supervised approaches alongside a LLaMA-based model to generate concise, plain-language summaries tailored to patient needs. Evaluation using readability metrics and qualitative faithfulness analysis shows substantial gains in accessibility while preserving essential clinical information. These results demonstrate that a unified LLM-based approach can reduce language barriers and better support patients in understanding and selecting clinical trials.

1 Introduction

Clinical trials are essential for advancing medical knowledge and patient care, but patient engagement depends on clear understanding of study goals and participation requirements. Although platforms such as ClinicalTrials.gov (4) aim to improve transparency, clinical trial descriptions often contain dense medical terminology and complex regulatory language that exceed recommended readability levels for patient-facing materials. As a result, potential participants may struggle to understand a study’s purpose, eligibility criteria, risks, and procedures, discouraging participation and limiting informed trial selection.

To address this challenge, we propose an automated, end-to-end pipeline for generating patient-friendly clinical trial descriptions. Our approach integrates summarization, linguistic simplification, medical jargon identification and explanation, and readability evaluation within a unified framework. Through extensive experimentation, we show that instruction-tuned large language models can produce concise, medically faithful summaries that are substantially easier to read and better support informed engagement with clinical research

2 Related Work

Improving the accessibility of clinical text has been studied across NLP and health informatics, including medical text simplification, biomedical summarization, and readability assessment. Rule-based simplification using consumer health vocabularies (1) can improve readability but may intro-

duce factual errors. Biomedical summarization has been widely applied to scientific and clinical texts (2; 3), yet transformer-based models typically retain technical language and are not optimized for patient audiences. Readability metrics such as Flesch–Kincaid (10) and Gunning Fog Index consistently show that clinical trial descriptions exceed recommended patient reading levels, motivating approaches that combine simplification with medical jargon identification and explanation.

This work builds on these research directions by integrating summarization, simplification, jargon identification, and readability evaluation into a unified pipeline tailored to patient facing clinical trial descriptions.

3 Dataset Overview

We use the *Clinical Trials* dataset released by Brulenaudet et al. and hosted on Hugging Face¹. The dataset is derived from publicly available records on ClinicalTrials.gov (4). Each record corresponds to a single registered clinical trial and includes a combination of structured metadata and free-text narrative fields.

The dataset contains administrative identifiers, structured attributes such as study phase and conditions, and multiple long form textual fields. For our task, the most relevant fields are the `brief_summary`, `eligibility_criteria` and `detailed_description`. The `brief_summary` is intended to provide a high level overview of the study, while the `detailed_description` offers a comprehensive account of the study’s purpose, design, and procedures. In practice, both fields frequently contain technical language and complex sentence structures, making them difficult for non expert readers.

3.1 Text Characteristics and Filtering

An analysis of text length reveals substantial variability across trials, with detailed descriptions often spanning several hundred words. This motivates the need for automatic summarization. We also observe missing or minimal text in certain fields for a subset of trials, which can negatively affect downstream modeling. To ensure reliable processing, we filter out trials lacking sufficient descriptive content, resulting in a high quality subset, Figure 1.

3.2 Readability and Complexity

Readability analysis indicates that clinical trial descriptions are written at advanced reading levels, often requiring college level literacy, and contain a high density of medical jargon and abbreviations. As a result, this dataset provides a realistic and challenging testbed for evaluating patient-friendly summarization and readability aware generation methods.

4 Methodology

4.1 Summarization Module - Trial

We first attempted to train a dedicated summarization model on the clinical trials dataset in order to compress long trial descriptions into short patient-oriented summaries. For this trial, we used our filtered dataset (saved to disk after milestone 2 filtering) and trained on a subset of 2000 examples for feasibility. Each example was converted into an instruction-style input that concatenates multiple trial fields: `detailed_description`, `brief_summary`, and `eligibility_criteria`. The input was lightly normalized (Unicode normalization, HTML removal, whitespace collapsing)

	field	num_missing	pct_missing
0	interventions	541897	100.000000
1	study_population	420133	77.530047
2	maximum_age	254447	46.954864
3	keywords	201556	37.194522
4	detailed_description	175870	32.454507
5	minimum_age	36025	6.647942
6	outcomes	17510	3.231241
7	healthy_volunteers	13771	2.541258
8	official_title	9986	1.842786
9	design_info	4664	0.860680
10	sex	1337	0.246726
11	eligibility_criteria	1014	0.187120
12	conditions	955	0.176233
13	study_type	920	0.169774
14	brief_summary	920	0.169774
15	nct_id	0	0.000000
16	overall_status	0	0.000000
17	brief_title	0	0.000000

Figure 1: Missing values distributions

¹<https://huggingface.co/datasets/louisbrulenaudet/clinical-trials>

and then truncated conservatively using a sentence-boundary heuristic (cut at the last “.” within a maximum character budget) to reduce the risk of breaking semantics.

To keep training lightweight, we used `google/flan-t5-small` as a sequence-to-sequence backbone. Inputs were tokenized to a maximum of 512 tokens and targets to 128 tokens. The model was trained using the HuggingFace `Seq2SeqTrainer` for 3 epochs with beam-search generation enabled during evaluation. The optimization setup used a small effective batch size (batch size 4 with gradient accumulation of 2) and periodic evaluation every 50 steps.

Training objective and limitation. In this trial setup, the `brief_summary` field was used as the “gold” target summary. This choice allowed immediate supervised training without manual annotations, but it introduced a critical limitation: many `brief_summary` entries are either too short, too technical, or stylistically inconsistent with patient-friendly language. In practice, the trained model tended to learn a near-copying behavior of the brief summary rather than producing a new, higher-quality abstraction from the detailed description.

Evaluation and observed failure mode. We conducted a short evaluation using ROUGE on the held-out validation split. Quantitatively, ROUGE could be computed, but qualitative inspection revealed the key issue: the model output often resembled the original `brief_summary` rather than a patient-centered summary of the full trial content. This confirmed that, under this training objective, the model was not learning the desired transformation. This outcome motivated our later pivot toward prompt-based LLM generation where the target style (patient-friendly and jargon-averse) could be explicitly controlled without relying on `brief_summary` as ground truth.

4.2 Simplification Module - Trial

After summarization, the next step in our pipeline is linguistic simplification: rewriting technical text into clearer, more readable language while preserving meaning. Unlike summarization, the clinical trials dataset does not provide parallel pairs of (technical $\rightarrow$ simplified) text suitable for supervised training. To prototype this step, we trained a simplification model on an external parallel simplification dataset and then tested whether it transfers to clinical trial language.

External training data. We used the GEM `cochrane-simplification` dataset, which contains pairs of technical and simplified medical text. This dataset is closer to our domain than general-purpose simplification corpora and provides direct supervision for rewriting: “Cochrane is an English dataset for paragraph-level simplification of medical texts” (9)

Model and preprocessing. We trained a T5-style model (`t5-base`) for simplification. Inputs were formatted with an instruction prefix (e.g., “simplify for better reading and understanding:”) to encourage controlled rewriting. Tokenization used a maximum input length of 256 tokens and maximum target length of 128 tokens. Training used `Seq2SeqTrainer` with ROUGE as an overlap metric against reference simplifications, and we additionally computed readability-oriented metrics on generated outputs: Flesch–Kincaid grade, Flesch Reading Ease, and an approximate jargon density based on Zipf word frequency thresholds.

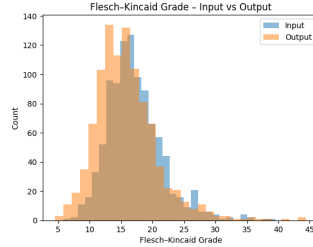


Figure 2: Flesch–Kincaid Grade of `brief_summary` and “simplified” `brief_summary`

Learning rate exploration. To improve stability and avoid under/over-training, we implemented a Leslie Smith-style learning-rate range test (LR finder) over a log-spaced schedule. The selected learning rate was guided by the region where loss decreased most steeply prior to divergence. Despite this tuning, training curves flattened quickly, and improvements were smaller than expected.

Observed limitation. To test applicability to our task, we loaded a subset (1000 rows) of the filtered clinical trials dataset and applied the trained simplifier to the `brief_summary` field. We prompted generation using a patient-oriented instruction (“Rephrase for better understanding from

patient point of view:”) and computed readability/jargon metrics before and after generation. Qualitative inspection showed that the model frequently produced outputs that were extremely similar to the input, often paraphrasing minimally rather than truly simplifying. This suggests that simplification learned from Cochrane-style text did not transfer strongly to the phrasing and structure of ClinicalTrials.gov summaries, and that a more direct approach (e.g., instruction-tuned LLM generation tailored to clinical trials) would be required. Readability scores also showed that the model wasn’t, in fact, simplifying the content of the text at all, Figure 2

4.3 Glossary Formation and Medical Term Explanation

Even after summarization and simplification, clinical trial descriptions inevitably contain medical terminology that cannot be removed without risking loss of meaning or factual accuracy. To address this issue, we introduce a glossary generation component that complements the patient-friendly summaries by explicitly explaining technical terms. Rather than attempting to replace medical vocabulary, this component aims to support comprehension by providing short, accessible explanations alongside the original terms.

4.3.1 Medical Term Identification

The first step in glossary formation is the identification of medically relevant terms within clinical trial text. Given the domain-specific nature of the data, a biomedical Named Entity Recognition (NER) approach is employed. Specifically, we use the `scispaCy (en_core_sci_lg)` model (5), which is trained on large-scale scientific and biomedical corpora and provides broader coverage than models restricted to diseases or chemicals alone. An earlier approach using a narrower biomedical NER model (`en_ner_bc5cdr_md`) (6) combined with frequency-based heuristics was explored, but this setup missed many clinically relevant concepts and introduced non-medical terms, making it unsuitable for reliable glossary construction.

In addition to entity recognition, abbreviation detection is incorporated to handle the frequent use of medical acronyms in clinical trial documentation. Both short forms and their corresponding expanded expressions are captured when available, ensuring that glossary entries remain interpretable for non-expert users.

To improve quality of the terms selected, a controlled fallback mechanism is applied for terms that are not identified as named entities. This mechanism operates at the single-word level and combines multiple signals, including word rarity statistics and rule-based morphological patterns commonly observed in biomedical terminology, such as characteristic prefixes, suffixes, and substrings. At the same time, a curated blacklist of non-medical and trial-related boilerplate terms is used to prevent common but non-informative words from being included. This hybrid strategy balances precision and coverage, enabling the glossary to capture meaningful medical concepts while limiting noise. All extracted terms are normalized to ensure consistent formatting and filtered to remove clearly invalid candidates, such as numeric expressions, measurement units, or very short tokens. This lightweight preprocessing step removes unsuitable spans early in the pipeline, allowing the medical classification logic to focus on semantically meaningful terms.

4.3.2 Definition Retrieval

For each identified medical term, a short explanation is retrieved from external knowledge sources. BioPortal (8) is used as the primary definition source, with the Medical Subject Headings (MeSH) ontology (7) selected due to its standardized terminology and widespread use in biomedical contexts. BioPortal provides structured, authoritative definitions that align well with clinical language and terminology.

When a definition cannot be found through BioPortal, Wikipedia is used as a fallback source. Although less formal, Wikipedia summaries are generally easier to read and more suitable for a lay audience, making them appropriate for patient-facing explanations. To improve efficiency and ensure reproducibility, all retrieved definitions are cached locally using a lightweight SQLite database, reducing repeated external requests when the same terms appear across multiple trial descriptions.

The final glossary is constructed by mapping each extracted medical term to its corresponding definition. For a given clinical trial description, the output is a structured collection of term–definition

pairs that can be displayed alongside the patient-friendly summary or accessed independently. An example from the final glossary looks like this: **”deferasirox”** = *A triazole and benzoate derivative that acts as a selective iron chelator. It is used in the management of chronic IRON OVERLOAD due to blood transfusion or non-transfusion dependent THALASSEMIA* or **”chelation”** = *Therapy of heavy metal poisoning using agents which sequester the metal from organs or tissues and bind it firmly within the ring structure of a new compound which can be eliminated from the body.*

4.4 LLM-Based Generation with Groq and LLaMA

Given the limitations of both supervised summarization (learning to replicate `brief_summary`) and external-data simplification transfer (minimal rewriting), we adopted a direct LLM-based generation approach as the primary method for producing patient-readable trial summaries. The key idea is to collapse summarization and simplification into a single controlled generation step: instead of training a model to match an imperfect reference, we instruct an LLM to produce a medically accurate, patient-friendly paragraph directly from structured and unstructured trial fields. For each trial, we build a single input by concatenating the most informative fields used throughout the notebook. This ensures the LLM has access to both the narrative description and the most patient-relevant constraints (who can join, age range, etc.).

Groq-hosted LLaMA generation. We used the Groq API with the instruction-tuned model `llama-3.3-70b-versatile`. The model is prompted to produce a single medically accurate paragraph in patient-friendly language and to avoid unnecessary jargon. We used a low temperature (0.2) to reduce variability and improve factual consistency, with a constrained output budget (typically 200–300 tokens). Naively summarizing one example per request is slow and fragile in real settings due to rate limits and network failures. To address this, we implemented a fault-tolerant batch pipeline with: (1) periodic checkpointing to a JSON progress file, (2) resumable execution from the last completed index, (3) exponential backoff retries when the API fails, and (4) structured output parsing. In particular, the LLM is asked to return a JSON array of objects with `{index, summary}` keys for each batch, and we validate that the parsed output matches this schema before accepting it. The final output is saved as a HuggingFace dataset on disk (e.g., a subset dataset containing a new `combined_summary` column).

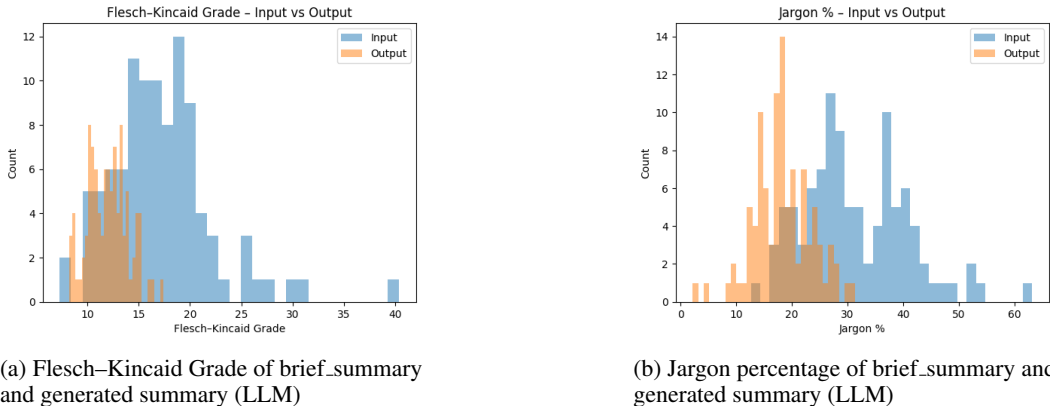


Figure 3: Readability metrics before and after LLM-based summarization/simplification.

Readability and jargon evaluation. To quantify accessibility improvements, we compute readability metrics and jargon density for both the original `brief_summary` (input baseline) and the generated `combined_summary` (LLM output). Using the same metric suite defined earlier in the notebook (Flesch–Kincaid grade, Gunning Fog, SMOG, Flesch Reading Ease, and Zipf-frequency-based jargon ratio), we observe substantial improvements on the Groq+LLaMA generated summaries compared to the original trial summaries, Figure 3. Qualitative inspection further confirms that the generated paragraphs integrate multiple trial fields while remaining easier to read and more coherent from a patient perspective, Table 1.

Patient-facing use case. A patient begins by selecting a clinical trial of interest from the dataset. Initially, they are presented with the original `detailed_description` and `brief_summary`, which are often difficult to understand due to length, complex sentence structure, and medical terminology. The system then provides a simplified patient-oriented summary generated by the LLM-based pipeline, offering a concise and clearer explanation of the trial’s purpose, target population, and participation requirements. Alongside the text, the interface displays quantitative readability indicators (e.g., Flesch–Kincaid Grade Level, Gunning Fog Index, and jargon density), allowing users to directly observe improvements in accessibility. Finally, the patient can request an automatically generated glossary, which extracts medical terms appearing in the trial description and provides short explanations, enabling deeper understanding without requiring external resources.

Initial brief summary	Generated simplified summary
The primary aim of this observational study is to investigate specific aspects of social cognition in dystrophinopathies. Body awareness, interpersonal distance and emotional processing will be measured in a sample of patients affected by Becker (BMD) and Duchenne (DMD) muscular dystrophy, compared with a sample of patients affected by osteogenesis imperfecta (OI), and both compared with a control sample with typical development. The secondary aim is to study cortical activity at rest, by means of electroencephalography (EEG), to explore frequencies and time course of EEG responses. Moreover, the relationship between EEG activity and neuropsychological, dispositional and subjective measures will be explored through correlational analyses.	This study aims to understand how people with certain muscle disorders, such as Duchenne and Becker muscular dystrophy, and a condition called osteogenesis imperfecta, interact with others and understand their own emotions and body. The study will compare these individuals with people who do not have these conditions. Participants will complete questionnaires and tasks, such as identifying emotions and measuring their comfort with personal space, and will wear a special headset to measure their brain activity while they rest. The goal is to learn more about social cognition, or how people understand and interact with others, in these conditions and to see if there are any differences in brain activity. The study is looking for participants with these conditions, as well as healthy volunteers, aged 4 and up.

Table 1: Comparison between technical and plain-language study summaries

Overall, this LLM-based strategy became our most effective approach in the project: it avoids reliance on noisy supervised targets, produces better-controlled patient-friendly language, and can be made scalable through robust batching and checkpointed execution.

5 Conclusion

This project presents an end-to-end, patient-centered pipeline for generating readable clinical trial summaries from public trial records. Using an instruction-tuned LLaMA-based model, the system unifies summarization, linguistic simplification, medical jargon handling, and readability assessment into a single workflow, producing coherent plain-language summaries while preserving factual consistency with the original descriptions.

We evaluate the system using standard readability metrics and qualitative faithfulness analysis, showing that the generated summaries are substantially easier to read while retaining essential information such as study purpose, target population, and participation requirements. These results demonstrate the potential of large language models to improve patient access to clinical research and support informed comparison and selection of clinical trials.

Future Work

Future work will focus on expanding and strengthening the evaluation of the proposed system. This includes more systematic faithfulness checks, human evaluation of patient comprehension. Additional improvements may involve incorporating structured metadata more explicitly into generation prompts, refining jargon explanations, and adapting the pipeline for multilingual or low literacy populations. Ultimately, this work aims to contribute toward more transparent, inclusive, and patient friendly access to clinical research.

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